

Name Brandon Buchanan



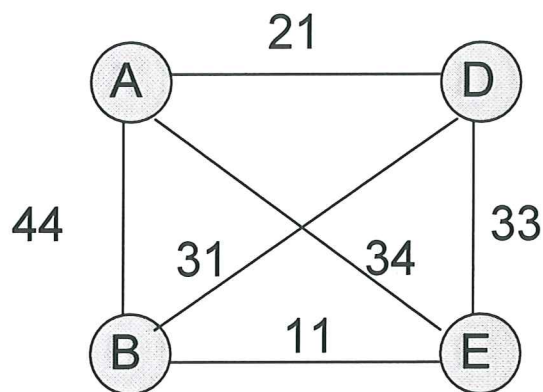
1) What is the optimal TSP?

A, E, D, B, A

2) Starting at node B, create a greedy algorithm to find a tour. Is it optimal?

B → E → D → A → B

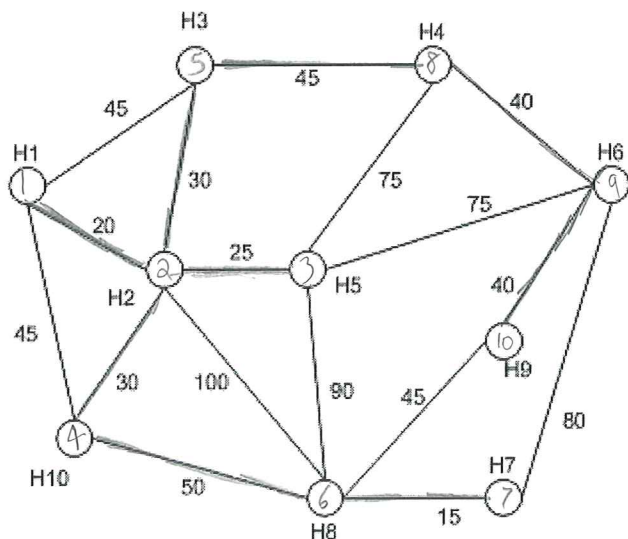
it is not optimal



3) 2-Approximation for Metric TSP

Find the MST

List the vertices in a preorder walk



1-3, 4-7, 1-4

4) 1.5 Approximation for Metric TSP TSP $D(u,v) \leq D(u,w) + D(w,v)$ (triangle inequality)

Find MST = T,

Let U be the odd vertices of T.

Find perfect matching of U in G (includes all vertices)

MUP is tour